

BTEC Level 3 Applied Science · Unit 1 Biology · Answer sheet

Microscopy — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. $M = \text{image size} / \text{actual size}$.
2. Smallest distance at which two points can still be seen as separate.
3. $2500\ \mu\text{m}$; $2.5 \times 10^6\ \text{nm}$.

B

4. Eyepiece magnification $\times$ objective magnification.
5. TEM: electrons through a thin specimen, internal detail. SEM: surface scan, 3D appearance.
6. Electron microscopes require a vacuum; living specimens cannot be used.

C

7. $18\ \text{mm} = 18\ 000\ \mu\text{m}$; $\text{actual} = 18\ 000/600 = 30\ \mu\text{m}$.
8. $3.0\ \text{cm} = 30\ 000\ \mu\text{m}$; $M = 30\ 000/7.5 = \times 4000$.
9. $40\ \text{mm} = 40\ 000\ \mu\text{m}$; $M = 40\ 000/10 = \times 4000$.

D

10. $35\ \text{mm} = 35\ 000\ \mu\text{m}$; $\text{actual} = 70\ \mu\text{m}$.
11. Light about $0.2\ \mu\text{m}$; TEM much smaller (about $0.1\ \text{nm}$) because electrons have a shorter wavelength.
12. Living specimens / colour / cheaper / easier preparation.

E

13. The granules are closer together than the resolution. Extra magnification enlarges the blur but cannot separate them (empty magnification).